

Highlights

Bounding the external field in mean-field opinion dynamics: what any content observable must satisfy to drive a transition

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- Media fields in Ising opinion models are usually assigned, not measured.
- For $h = \alpha X$, a spinodal crossing needs $|\alpha| \geq h_c(\beta J)/\Delta X$.
- Closed-form spinodal field, $h_c = 0.5328$ at $\beta J = 2$.
- Landau coefficients derived; mean-field exponents recovered numerically.
- A necessary-condition screen for content observables, run before data.

Bounding the external field in mean-field opinion dynamics: what any content observable must satisfy to drive a transition

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Abstract

Mean-field opinion models from the Ising universality class are routinely used to argue that media content can tip a population between competing consensus states. In almost every case the external field representing that content is assigned by hand, so the models show that *a* field of sufficient magnitude tips the population while saying nothing about whether any real content observable reaches that magnitude. We address that gap from the theory side. Writing the field as $h = \alpha X$ for a content observable X whose attainable range ΔX contains its neutral value, we show that in all-to-all mean-field theory, starting from the symmetric zero-field state, no media-driven transition across the spinodal is possible unless $|\alpha| \geq h_c(\beta J)/\Delta X$, where $h_c(\beta J)$ is the spinodal field at which the metastable state disappears. The spinodal field has the closed form $h_c = \beta J m_s - \text{artanh}(m_s)$ with $m_s^2 = 1 - 1/(\beta J)$, giving $h_c = 0.5328$ at $\beta J = 2$. The condition is necessary, not sufficient, and it does not measure α : it states how large the coupling would have to be for the mechanism to operate at all, and it can be evaluated before any outcome data are collected. We derive the Landau coefficients from the self-consistency condition rather than asserting them, recover the mean-field exponents $\beta = 1/2$, $\gamma = 1$ and $\delta = 3$ numerically as a check on the solver, and state a screening criterion that any proposed field observable must pass.

Keywords: sociophysics, opinion dynamics, mean-field Ising model, Landau theory, spinodal, external field

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1. Introduction

The statistical physics of social dynamics has a standard move: identify a binary attitude with an Ising spin, identify social conformity with a ferromagnetic coupling J , identify media or propaganda with an external field h , and study when the population’s average opinion m jumps between consensus states [1, 2, 3, 4]. The move is productive. It supplies an order parameter, a control parameter and a transition, and it explains why small changes in influence can produce disproportionate changes in collective opinion near a critical point.

It also has a weakness that is rarely stated. The field h enters as a free parameter. A study fixes h at whatever value makes the phenomenon visible, sweeps it, and reports that the population tips at some h^* . The pattern is common in the literature that models media explicitly: Crokidakis [5] introduces mass media into the two-dimensional Sznajd model as a probability p of following the media opinion and sweeps it, finding the usual phase transition absent for $p \gtrsim 0.18$; Azhari and Muslim [6] apply an external field to majority-rule and q -voter dynamics on the complete graph and sweep its strength; Freitas, Vieira and Anteneodo [7] show in a kinetic-exchange model that an external bias gives rise to imperfect bifurcations and hysteresis as its strength is varied; and the review by Sîrbu et al. [8] surveys models that combine peer interaction with external information, in which the external term is specified as a model parameter. In each of these studies the swept parameter is the quantity whose real-world magnitude is at issue.

Nothing in such an exercise establishes that real media content produces a field anywhere near h^* , because nothing in it connects h to a measurable property of content. The conclusion “a sufficiently strong field tips the population” is then a statement about the model, not about media. The recent review of Ising-inspired sociophysics by Mullick and Sen [9] points the same way, listing data-driven calibration against large-scale social datasets among the promising directions for future work.

One recent study breaks the pattern and is worth stating precisely, because it bounds what this paper can claim. Korb, Dahdoul and Thurner [10] model elections as a random field Ising model with a bimodal field representing two campaigns, calibrate it to historical US House election data, and extract a critical campaign spending. Their field is measured, not assigned,

which is the move this paper argues for.

Two things nonetheless remain open. Their calibration is fitted from outcome data after the fact, so it cannot screen a proposed mechanism before the data exist; and it is specific to expenditure, whereas observables proposed for media *content* remain uncalibrated. What is missing in both cases is a necessary condition stated on the coupling itself.

This paper takes that step in the direction that does not require outcome data. Suppose an observable X is proposed as the content-side quantity driving the field, through a linear map $h = \alpha X$ with an unknown coupling α . Suppose further, as is true of any real observable, that X has a bounded attainable range ΔX across the content a population actually encounters. Then, within all-to-all mean-field theory and for a population starting from the symmetric zero-field state, the requirement that the field be able to reach the spinodal imposes a lower bound on the coupling that involves no fitting:

$$|\alpha| \geq \frac{h_c(\beta J)}{\Delta X}. \quad (1)$$

Everything on the right is either a property of the model (h_c as a function of the coupling βJ) or a property of the observable that can be measured without outcome data (ΔX). The bound is therefore a screening test: it can reject a proposed mechanism, within the model, before any expensive measurement of α is attempted, and it converts a failure to detect a coupling into a quantitative statement about how large the coupling would have had to be. It is a necessary condition only; it is not a prediction that media content moves opinion, and it does not measure α .

Section 2 fixes the model and derives the Landau coefficients from the self-consistency condition. Section 3 validates the solver against the exact mean-field exponents, gives the spinodal field in closed form and derives the bound. Section 4 states the screening criterion and the scope of the result, and Section 5 concludes.

2. Model

2.1. Mean-field Ising model of opinion

Take N agents carrying binary opinions $s_i = \pm 1$, coupled all-to-all with strength J/N , in an external field H :

$$\mathcal{H} = -\frac{J}{N} \sum_{i < j} s_i s_j - H \sum_i s_i. \quad (2)$$

The magnetisation $m = N^{-1} \sum_i s_i$ is the collective opinion polarisation. With $\beta = 1/k_B T$, $k_B = 1$, and the reduced field $h \equiv \beta H$, it obeys in the limit $N \rightarrow \infty$ the self-consistency condition

$$m = \tanh(\beta J m + h). \quad (3)$$

The reduced field h is dimensionless and directly comparable to βJ ; it is the field used in the rest of the paper. The social reading of β is the inverse of a noise level: large β describes a population that copies others reliably, small β one that decides close to at random.

At zero field, Eq. (3) has the single root $m = 0$ for $\beta J \leq 1$ and three roots for $\beta J > 1$, of which the outer two are stable. The transition at $\beta J = 1$ is continuous.

2.2. Landau expansion with derived coefficients

Write the free energy per agent, in units of $k_B T$, as a quartic in the order parameter,

$$F(m) = a m^2 + b m^4 - h m. \quad (4)$$

The coefficients are not free. Requiring that $\partial F / \partial m = 0$ reproduce Eq. (3) to cubic order fixes them. Inverting Eq. (3) gives $\text{artanh}(m) = \beta J m + h$, and expanding $\text{artanh}(m) = m + m^3/3 + O(m^5)$ yields

$$(1 - \beta J) m + \frac{1}{3} m^3 - h = 0. \quad (5)$$

Matching term by term against $\partial F / \partial m = 2am + 4bm^3 - h$ gives

$$a = \frac{1 - \beta J}{2}, \quad b = \frac{1}{12}. \quad (6)$$

The same coefficients follow from expanding the mean-field (Bragg–Williams) free energy $-\frac{1}{2} \beta J m^2 - h m + \frac{1+m}{2} \ln \frac{1+m}{2} + \frac{1-m}{2} \ln \frac{1-m}{2}$ to fourth order, which equals $-\ln 2 + F(m) + O(m^6)$, so the two routes are consistent. We state the derivation because a quartic with coefficients that do not follow from the accompanying self-consistency condition still has a qualitatively correct double well while its minimum sits away from the root m^* of Eq. (3); the error is easy to miss in a figure. The double well opens when $a < 0$, that is when $\beta J > 1$, recovering the transition already implied by Eq. (3). Figure 1 shows Eq. (4) at zero field on both sides of the critical coupling.

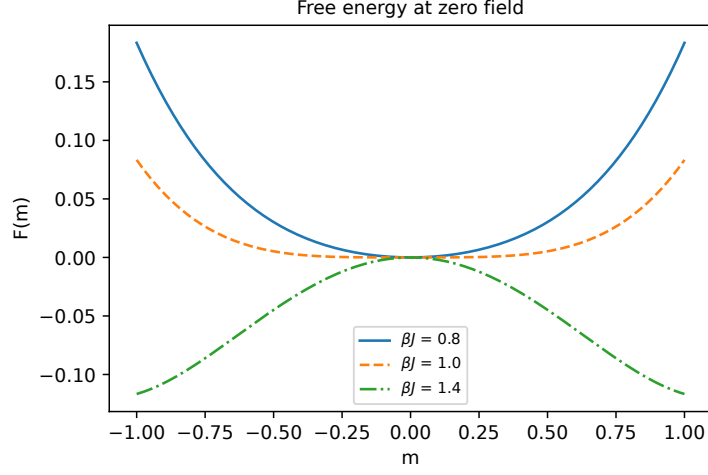


Figure 1: Free energy, Eq. (4), at zero field for $\beta J = 0.8, 1.0$ and 1.4 with the coefficients of Eq. (6). The double well appears only above $\beta J = 1$.

The quartic is used here for the qualitative structure and the exponents. The spinodal field of Section 3.2 is computed from the full Eq. (3), not from the truncated quartic, which reproduces it only as $\beta J \rightarrow 1^+$.

The linear response to the reduced field is

$$\chi \equiv \frac{\partial m}{\partial h} = \frac{\text{sech}^2(\beta J m^* + h)}{1 - \beta J \text{sech}^2(\beta J m^* + h)}, \quad (7)$$

so that the response to the unreduced field is $\partial m / \partial H = \beta \chi$. It diverges where the denominator vanishes; we report χ only on stable roots, where the denominator is positive.

3. Results

3.1. Numerical validation against the exact exponents

Every later figure is drawn with the same solver, so the solver is checked first against results that are known exactly. Mean-field theory gives [11]

$$\beta = \frac{1}{2}, \quad \gamma = 1, \quad \delta = 3, \quad (8)$$

for $m^* \sim (\beta J - 1)^\beta$ at zero field, $\chi \sim |1 - \beta J|^{-\gamma}$, and $m \sim h^{1/\delta}$ on the critical isotherm $\beta J = 1$. (The exponent β is not to be confused with the

inverse temperature; context distinguishes them.) All three follow directly from Eq. (5): $m^{*2} = 3(\beta J - 1)$ at zero field, $\chi = 1/(1 - \beta J)$ at $m^* = 0$, and $m^3 = 3h$ at $\beta J = 1$.

One methodological point matters. Solving Eq. (3) by direct iteration of $m_{k+1} = \tanh(\beta J m_k + h)$ is the natural implementation and is adequate away from criticality, but its convergence degrades at $\beta J = 1$, where the map's derivative reaches unity. The exponents are defined in that limit, so fitting them to iterated values measures truncation error rather than physics: with 10^3 iterations and tolerance 10^{-10} , doing so returns $\delta \approx 1.35$ against an exact 3. We therefore fit the exponents to roots obtained by bracketing, and check the iterative solver against the bracketed one away from criticality ($|\beta J - 1| \geq 0.02$), where the largest absolute discrepancy over the sampled grid is 1.15×10^{-9} .

Fitting β and γ at 60 points in a window of width 0.05 in $|\beta J - 1|$, and δ at 60 log-spaced fields $10^{-6} \leq h \leq 10^{-3}$, recovers

$$\beta = 0.4872, \quad \gamma = 1.0000, \quad \delta = 3.0042, \quad (9)$$

with absolute errors 0.0128, $< 10^{-14}$ and 0.0042 respectively. The γ fit is exact by construction, since $m^* = 0$ below the transition at zero field and $\chi = 1/(1 - \beta J)$ identically; it checks the susceptibility expression and the fitting procedure rather than the root finder. The residual error in β is a correction to scaling: keeping the next term of the expansion, $m^5/5$, gives $m^{*2} = 3t(1 - \frac{9}{5}t + O(t^2))$ with $t = \beta J - 1$, which lowers the effective log-log slope across a finite window. Figures 2, 3 and 4 show the three fits.

3.2. The spinodal field

Above $\beta J = 1$ the free energy has two minima at zero field. A field deepens one and shallows the other; at a finite field the shallower minimum disappears. That value is the spinodal field $h_c(\beta J)$: the field a population in the metastable state must experience before, within mean-field theory, it is obliged to move.

It follows in closed form from Eq. (3). Write the roots as solutions of $h = g(m) \equiv \text{artanh}(m) - \beta J m$. A pair of roots merges and disappears where $g'(m) = 1/(1 - m^2) - \beta J = 0$, that is at

$$m_s^2 = 1 - \frac{1}{\beta J}, \quad (10)$$

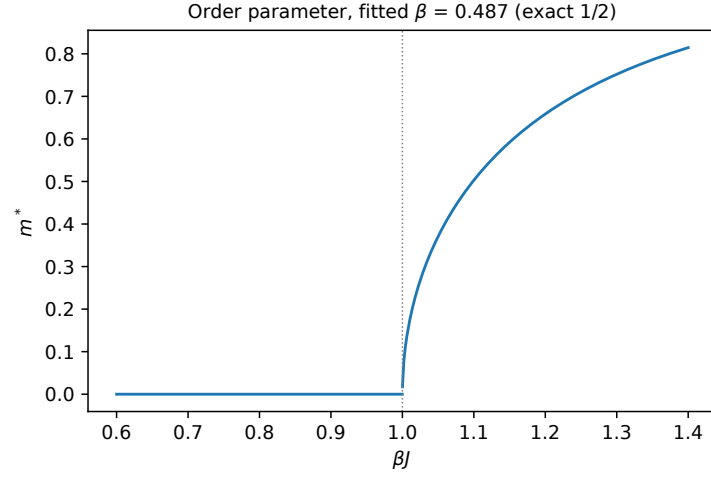


Figure 2: Order parameter across the transition. The fitted exponent is 0.4872 against an exact $1/2$.

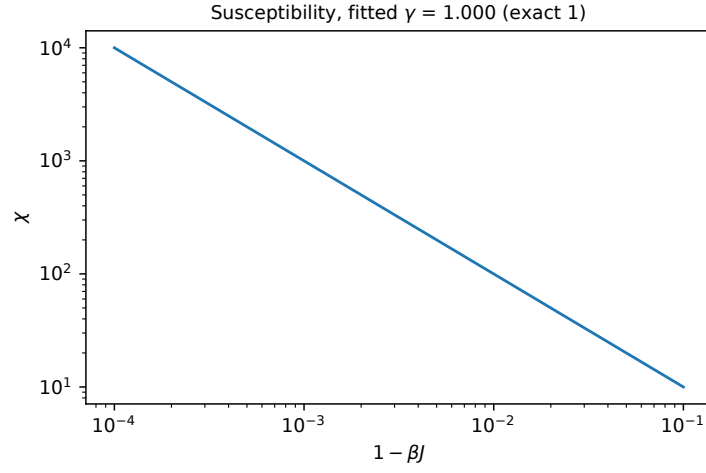


Figure 3: Susceptibility, Eq. (7), approaching criticality from below at zero field; fitted exponent 1.0000 against an exact 1.

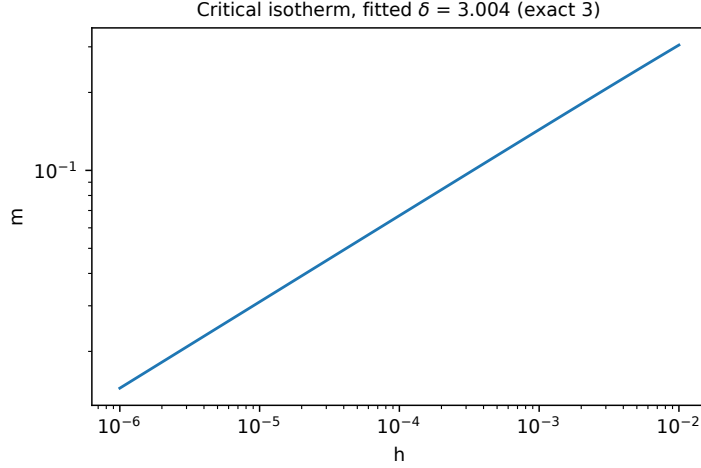


Figure 4: Critical isotherm at $\beta J = 1$, fitted exponent 3.0042 against an exact 3.

and the corresponding field is

$$h_c(\beta J) = |\operatorname{artanh}(m_s) - \beta J m_s| = \beta J m_s - \operatorname{artanh}(m_s). \quad (11)$$

At $\beta J = 2$, $m_s = 1/\sqrt{2}$ and $h_c = \sqrt{2} - \operatorname{artanh}(1/\sqrt{2}) = 0.53284$. As $\beta J \rightarrow 1^+$, Eq. (11) gives $h_c \simeq \frac{2}{3}(\beta J - 1)^{3/2}$, the value the quartic of Eq. (4) predicts; away from the transition the quartic overestimates it (2/3 against 0.53284 at $\beta J = 2$). The figures are drawn from a bisection on the number of stable roots of Eq. (3), which agrees with Eq. (11) to better than 10^{-6} over the plotted range. Figure 5 shows the boundary; h_c vanishes as $\beta J \rightarrow 1^+$ and increases monotonically with βJ .

3.3. The bound

Now let the field be generated by a content observable. Any such proposal has the form

$$h = \alpha X, \quad (12)$$

where X is measurable on a piece of content and α is an unknown coupling carrying the units. Real observables are bounded: across the content a population plausibly encounters, X lies in some interval $[X_{\min}, X_{\max}]$ of width $\Delta X = X_{\max} - X_{\min}$, which is an empirical property of the observable and of the corpus, not of the model.

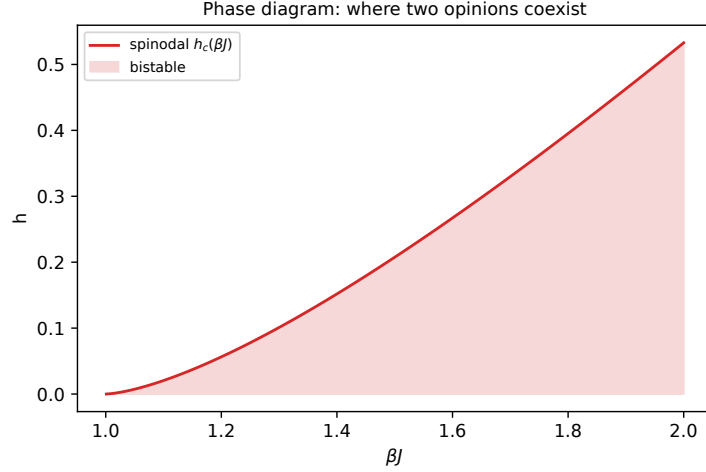


Figure 5: Phase diagram in $(\beta J, h)$ for $h \geq 0$; the diagram is symmetric under $h \rightarrow -h$. The shaded region is where two stable opinion states coexist; its upper boundary is the spinodal $h_c(\beta J)$ of Eq. (11).

Two further assumptions fix the reference point. First, the field is measured from the content-free reference $h = 0$, at which the population sits in one of the two symmetric states $\pm m^*$. Second, X is signed about a neutral value $X = 0$ that lies inside the attainable range, $X_{\min} \leq 0 \leq X_{\max}$, as for an observable defined as a difference between two opposing components. Then every attainable field satisfies $|h| \leq |\alpha| \max(|X_{\min}|, |X_{\max}|) \leq |\alpha| \Delta X$. Moving a population from either state into the other through a spinodal crossing requires a field of magnitude at least $h_c(\beta J)$ opposing that state, so a necessary condition is

$$|\alpha| \Delta X \geq h_c(\beta J) \quad \Longleftrightarrow \quad |\alpha| \geq \frac{h_c(\beta J)}{\Delta X}. \quad (13)$$

Replacing ΔX by $\max(|X_{\min}|, |X_{\max}|)$ gives a tighter condition; the form in Eq. (13) is the weaker one and so remains necessary under the stated assumptions. Figure 6 plots the right-hand side against βJ for several values of ΔX . At $\beta J = 2.00$ the required coupling is 5.3284 for an observable of spread 0.1, 1.0657 for spread 0.5, 0.5328 for unit spread and 0.2664 for spread 2.

Three properties of Eq. (13) are worth drawing out.

First, it is a necessary condition, not a sufficient one. Clearing the bound does not establish that content moves populations; it only fails to rule it out.

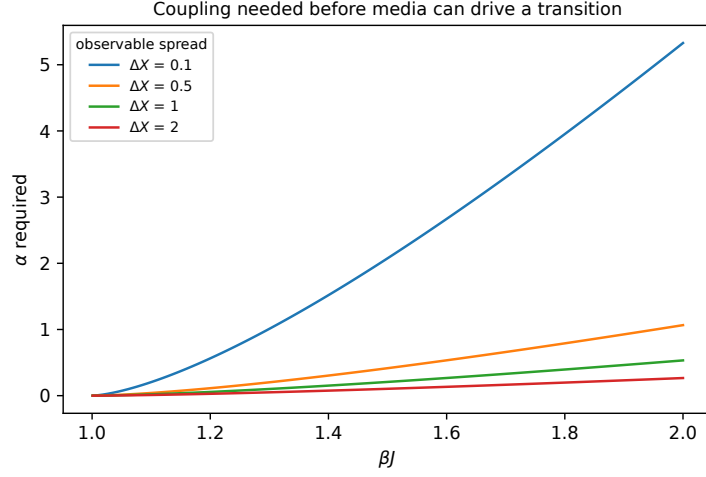


Figure 6: Coupling $h_c(\beta J)/\Delta X$ required before a content observable of spread ΔX can drive the population across the spinodal, Eq. (13).

Falling below it excludes the mechanism within the model’s assumptions, for a population that must cross the spinodal.

Second, it does not require the coupling to be estimated. ΔX is obtained by measuring the observable across a corpus, which does not need the outcome variable and is ordinarily far cheaper than estimating a coupling to collective behaviour, and h_c comes from the model.

Third, it turns a null result into a quantitative constraint. A study that estimates α and obtains an interval containing zero has, on its own, only the statement that no coupling was detected. Combined with Eq. (13), the same interval states that the coupling would have to exceed $h_c(\beta J)/\Delta X$ for the mechanism to operate, and whether the measurement excludes that range at the stated confidence. This is available to any study that reports a spread alongside its estimate.

As an illustration, a companion study by the author (manuscript in preparation) measures a signed neural-response observable on a 400-article corpus and finds a range that contains zero, with spread $\Delta X = 0.1241$. Equation (13) then requires $|\alpha| \geq 0.53284/0.1241 \approx 4.29$ at $\beta J = 2$. This is a statement about what the coupling would have to be, not a measurement of it.

4. Discussion

4.1. A screening criterion

The bound supports a procedure that can be run before outcome data are collected.

1. State the observable X and the map $h = \alpha X$ explicitly, including units, and the neutral value $X = 0$.
2. Measure or bound ΔX on a sample of the content of interest, and check that the range contains the neutral value. This does not require the outcome variable.
3. Choose the range of βJ the population plausibly occupies and evaluate $h_c(\beta J)$ from Eq. (11).
4. Evaluate $\alpha_{\min} = h_c(\beta J)/\Delta X$ and ask whether a coupling of that size is defensible on independent grounds.
5. If it is not, the mechanism is excluded within the model and the expensive measurement is unnecessary. If it is, proceed, and report ΔX alongside the estimate of α so that the bound can be applied to the result.

The weakest link is step 4, since α has no independent scale in most applications. The criterion is still informative in relative terms: candidate observables can be ranked by $\alpha_{\min}$, provided they are first expressed on a common scale (for example, each standardised by its own standard deviation across the corpus), since rescaling an observable by a constant rescales $\alpha_{\min}$ by its inverse. An observable whose spread demands an implausibly large coupling is a poor choice of field regardless of how well it correlates with anything.

4.2. Scope and limitations

The result inherits every assumption of the all-to-all mean-field model: a single homogeneous population, a static field, binary opinions, and the thermodynamic limit. On sparse or finite-dimensional networks, with heterogeneous susceptibility, or with temporal dynamics, the switching field differs from Eq. (11), so Eq. (13) is a bound within this model rather than a universal one. The linear map $h = \alpha X$ is the simplest possible coupling and is an assumption in its own right; a saturating map would change the relation between ΔX and the maximum attainable field. The reference-point assumptions matter too: if the content-free field is not zero but already favours the

target state by $h_0 > 0$, the requirement relaxes to $|\alpha|\Delta X \geq h_c - h_0$, and if the attainable range of X does not contain its neutral value the maximum field is no longer bounded by $|\alpha|\Delta X$. Finally, the spinodal is the relevant threshold for a population trapped in a metastable state. In the all-to-all model at $N \rightarrow \infty$ the metastable state does not decay below h_c , but a finite population, or one on a network, can escape by activated nucleation at smaller fields given enough time; Eq. (13) then demands more of α than a nucleation argument would, and can overstate the coupling required. The bound does not measure α and does not predict that media content moves opinion.

5. Conclusion

In all-to-all mean-field Ising opinion dynamics with a content-driven field $h = \alpha X$, a population starting from the symmetric zero-field state cannot be carried across the spinodal by content unless $|\alpha| \geq h_c(\beta J)/\Delta X$, with h_c given in closed form by Eq. (11). The condition is necessary, not sufficient; it needs only the measured spread of the observable and the model’s spinodal, so it can screen a proposed mechanism before outcome data exist, and it turns a null estimate of the coupling into a statement about the range that estimate excludes. Extending the bound to networked, heterogeneous and finite populations, where nucleation replaces the sharp spinodal, is the natural next step.

Data availability

No empirical data were used; the paper is analytic and numerical. All figures and every number quoted are produced by `services/inference/scripts/phase_boundary.py`, which also enforces the exponent check (it exits non-zero if any fitted exponent departs from its exact value by more than a stated tolerance), and are written into the manuscript by `services/inference/scripts/paper1_numbers.py` rather than transcribed. The code and the generated records are public at <https://github.com/brn-mwai/monarch>.

CRediT authorship contribution statement

Brian Mwai: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Visualization, Writing – original draft, Writ-

ing – review & editing.

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